



7532 - A Joint Mid-IR and X-ray Investigation of the Physics Driving Sgr A*'s Flares

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRIstare				
	1	sgrAstare1	MIRI Medium Resolution Spectroscopy	(1) SgrA
	2	sgrAstare2	MIRI Medium Resolution Spectroscopy	(1) SgrA
	3	sgrAstare3	MIRI Medium Resolution Spectroscopy	(1) SgrA
	4	sgrAstare4	MIRI Medium Resolution Spectroscopy	(1) SgrA

ABSTRACT

The Galactic Center offers the closest opportunity for studying accretion onto a supermassive black hole. Flares from Sgr A* are detected across the electromagnetic spectrum and may originate in the accretion flow or the jet. Recent models indicate that this variability can be produced by a tilted inner disk, gravitational lensing of bright spots in the disk by the black hole, or particle acceleration in reconnection events. These models produce different flare characteristics, and better characterization of flares' spectral energy distributions will enable us to distinguish between emission models. Meanwhile, the circumnuclear environment of Sgr A* contains dozens of IR sources and structures that are critical to understanding both the accretion-feeding processes and the nature of the nuclear star cluster and nearby dusty gas features. Only JWST's MIRI has the high angular resolution and mid-IR sensitivity to probe this complex and dynamic region. MIRI can detect changes in the mid-IR spectral index, which is an important diagnostic of physical conditions in the flare. The X-ray flux will constrain the energy distribution of non-thermal particles in the flare. Thus, we propose MIRI MRS IFU time-series together with Chandra X-ray monitoring of Sgr A* (and its surroundings) to determine the nature of its flare emission and to test models of the accretion and radiation processes near our closest supermassive black hole.

OBSERVING DESCRIPTION

The goal of this program is to obtain time-series MIRI MRS IFU spectra of the region centered on Sgr A* in order to determine the spectra of the time-variable emission from the region near the supermassive black hole. These observations will be coordinated simultaneous with Chandra X-ray telescope time series observations.

We will use staring mode at the shortest MIRI MRS grating setting to monitor Sgr A* and detect fluctuations in the signal. This is broken into four separate observations of 25 ks (~7 hours) each, due to constraints on the length of staring observations of Sgr A* from Chandra. We will use instrument parameters similar to previous successful MIRI observations of Sgr A* flares: 30 groups/integration, and one exposure of 295 integrations for a total science time of ~25 ks for each exposure. Based on previous ground-based and Spitzer observations, we expect that during the monitoring

JWST Proposal 7532 (Created: Friday, August 1, 2025, 5:00:13PM Eastern Standard Time) - Overview

period there will be one peak or flare with an emission level of ~ 5 mJy at 5 microns, several peaks at the 2 mJy level, and many smaller fluctuations that we will be able to detect, at least at the shorter wavelengths, at higher S/N than previous observations.

The data will be collected in contiguous 25 ks (7 hr) exposures, rather than breaking it into smaller parts to avoid a High Gain Antenna movement during the exposure. This is preferable to separate exposures, which result in a ~ 30 min gap between data collection events.

We will also obtain simultaneous submm observations from ALMA and/or the SMA.

Proposal 7532 - Targets - A Joint Mid-IR and X-ray Investigation of the Physics Driving Sgr A*'s Flares

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SgrA	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0 Parallax: 0" Epoch of Position: 2000	
Comments: Category=Star Description=[Black holes] Extended=YES					

Proposal 7532 - Observation 1 - A Joint Mid-IR and X-ray Investigation of the Physics Driving Sgr A*'s Flares

Observation	Proposal 7532, Observation 1: sgrAstore1													Fri Aug 01 22:00:13 GMT 2025
	Diagnostic Status: Warning													
Observing Template: MIRI Medium Resolution Spectroscopy														
Diagnostics	(sgrAstore1 (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.													
	(sgrAstore1 (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.													
	(Visit 1:1) Warning (Form): Data Excess over middle threshold													
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	SgrA	RA: 17 45 40.0360 (266.4168167d)					Proper Motion RA: 0						
			Dec: -29 00 28.17 (-29.00783d)					Proper Motion Dec: 0						
			Equinox: J2000					Parallax: 0"						
								Epoch of Position: 2000						
	Comments: Category=Star Description=[Black holes] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	295	1	None	1	295	25374.966		
	1	SHORT(A)	MRSSHORT		FASTR1	30	295	1	None	1	295	25374.966		
Special Requirements	Time Series Observation No Parallel Attachments													

Proposal 7532 - Observation 2 - A Joint Mid-IR and X-ray Investigation of the Physics Driving Sgr A*'s Flares

Observation	Proposal 7532, Observation 2: sgrAstore2													Fri Aug 01 22:00:13 GMT 2025
	Diagnostic Status: Warning													
Observing Template: MIRI Medium Resolution Spectroscopy														
Diagnostics	(sgrAstore2 (Obs 2)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.													
	(sgrAstore2 (Obs 2)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.													
	(Visit 2:1) Warning (Form): Data Excess over middle threshold													
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	SgrA	RA: 17 45 40.0360 (266.4168167d)					Proper Motion RA: 0						
			Dec: -29 00 28.17 (-29.00783d)					Proper Motion Dec: 0						
			Equinox: J2000					Parallax: 0"						
								Epoch of Position: 2000						
	Comments: Category=Star Description=[Black holes] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	295	1	None	1	295	25374.966		
	1	SHORT(A)	MRSSHORT		FASTR1	30	295	1	None	1	295	25374.966		
Special Requirements	Time Series Observation No Parallel Attachments													

Proposal 7532 - Observation 3 - A Joint Mid-IR and X-ray Investigation of the Physics Driving Sgr A*'s Flares

Observation	Proposal 7532, Observation 3: sgrAstore3												Fri Aug 01 22:00:13 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(sgrAstore3 (Obs 3)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.													
	(sgrAstore3 (Obs 3)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.													
	(Visit 3:1) Warning (Form): Data Excess over middle threshold													
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	SgrA	RA: 17 45 40.0360 (266.4168167d)					Proper Motion RA: 0						
			Dec: -29 00 28.17 (-29.00783d)					Proper Motion Dec: 0						
			Equinox: J2000					Parallax: 0"						
								Epoch of Position: 2000						
Acquisition	Comments: Category=Star Description=[Black holes] Extended=YES													
	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	295	1	None	1	295	25374.966		
	1	SHORT(A)	MRSSHORT		FASTR1	30	295	1	None	1	295	25374.966		
Special Requirements	Time Series Observation No Parallel Attachments													

Proposal 7532 - Observation 4 - A Joint Mid-IR and X-ray Investigation of the Physics Driving Sgr A*'s Flares

Observation	Proposal 7532, Observation 4: sgrAstore4													Fri Aug 01 22:00:13 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(sgrAstore4 (Obs 4)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.													
	(sgrAstore4 (Obs 4)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.													
	(Visit 4:1) Warning (Form): Data Excess over middle threshold													
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	SgrA	RA: 17 45 40.0360 (266.4168167d)					Proper Motion RA: 0						
			Dec: -29 00 28.17 (-29.00783d)					Proper Motion Dec: 0						
			Equinox: J2000					Parallax: 0"						
								Epoch of Position: 2000						
	Comments: Category=Star Description=[Black holes] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	295	1	None	1	295	25374.966		
	1	SHORT(A)	MRSSHORT		FASTR1	30	295	1	None	1	295	25374.966		
Special Requirements	Between Dates 11-AUG-2025:18:30:00 and 11-AUG-2025:21:00:00 Time Series Observation No Parallel Attachments													